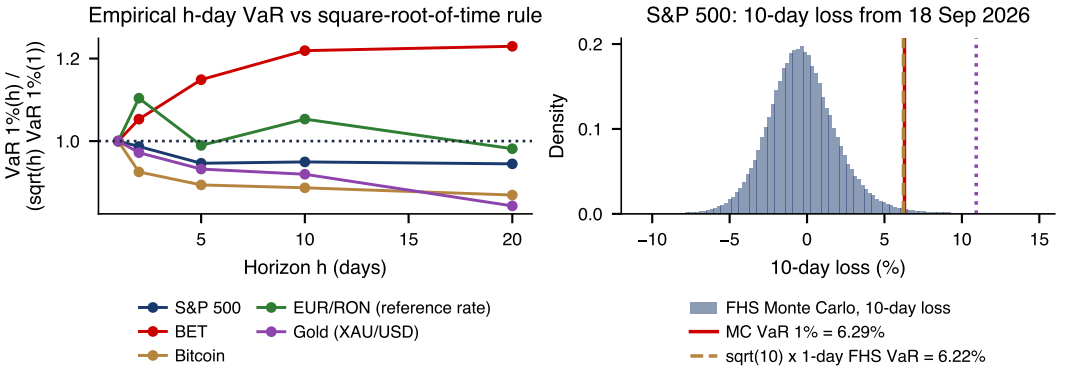
Empirical h-day VaR vs square-root-of-time rule
VaR 1%(h) / (sqrt(h) VaR 1%(1))
1.2
1.0
5
10
15
20
Horizon h (days)
S&P 500
BET
Bitcoin
EUR/RON (reference rate)
Gold (XAU/USD)
S&P 500: 10-day loss from 18 Sep 2026
Density
0.2
0.1
0.0
−10
−5
0
5
10
15
10-day loss (%)
FHS Monte Carlo, 10-day loss
MC VaR 1% = 6.29%
sqrt(10) x 1-day FHS VaR = 6.22%
sqrt(10) x 1-day HS VaR = 10.92%